ADS2001 Final Report

Portfolio Optimisation Using S&P 500 Stocks

Tutor/lecturer: Dr Hassan Fallahgoul
Jeffery Liu

Group members:
Angela Ngui
Tabriz Zarek
Jaek-Hxiang Leng
Kar Li Lee
Harjas Kaur

Table of Contents

No.	Title	Pages
	Executive Summary	3-4
	Glossary	4-5
1	**Introduction & Background Information** 1.1 Background to Portfolio Optimisation 1.2 Project Objectives 1.3 Description of the S&P 500 Dataset 1.4 Research Questions and Modelling Challenges 1.5 Overview of Methodology	5-6
2	**Data Quality and Exploratory Data Analysis** 2.1 Data Cleaning 2.2 Descriptive Statistics 2.3 Exploratory Data Analysis	6-8
3	**Model Development** 3.1 Portfolio Construction Framework 3.2 Minimum Variance Portfolio 3.3 Maximum Return Portfolio 3.4 Maximum Sharpe Portfolio 3.5 Machine Learning Models 3.6 Monte Carlo Simulation	8-15
4	**Results and Discussion** 4.1 Minimum Variance Portfolio Results 4.2 Maximum Return Portfolio Results 4.3 Maximum Sharpe Portfolio Results 4.4 Machine Learning Results 4.5 Monte Carlo Results 4.6 Portfolio Comparison	16-22
5	**Conclusion** 5.1 Key Findings 5.2 Recommended Portfolio Strategies 5.3 Limitations 5.4 Future Improvements	22-23
6	**References**	25-27
7	**Appendix**	28-32

Executive Summary

Background and Objective

This report addresses a fundamental investment question: given an initial capital of $100,000, how should funds be allocated across S&P 500 stocks to achieve the best balance between return and risk? To answer this, approximately 26 years of daily adjusted closing prices (September 1993 to July 2019) were analysed across three portfolio construction strategies — Minimum Variance, Maximum Return, and Maximum Sharpe Ratio — each targeting a distinct investment objective. Machine learning timing overlays and Monte Carlo simulation were additionally employed to assess whether adaptive allocation and random portfolio sampling could improve upon traditional optimisation methods.

Data

The analysis drew on daily adjusted closing prices for 1,199 US-listed securities, reduced to 453 investable stocks following the removal of warrants, distressed securities, and those with insufficient price history. An 80/20 temporal train-test split was applied, with all model parameters estimated on data from September 1993 to January 2013 and performance evaluated on the unseen period from January 2013 to July 2019.

Strategies and Results

Three portfolio strategies were constructed and evaluated. Their out-of-sample performance is summarised in Table 1.

Table 1. Out-of-Sample Portfolio Performance Summary

Strategy	Annualised Return	Annualised Volatility	Sharpe Ratio	No. of Stocks
Minimum Variance	7.60%	11.13%	0.682	19
Maximum Return (optimised)	12.05%	11.78%	1.023	10
Maximum Sharpe Ratio	19.76%	14.81%	1.200	32
Equal-Weight Benchmark	11.37%	16.36%	0.573	453

The **Minimum Variance Portfolio** achieved the lowest volatility (11.13%) by concentrating capital in 19 defensive stocks across Utilities, Consumer Staples, and Health Care. It generated a total return of approximately 143% over the test period. While this represents the most conservative outcome, its Sharpe ratio of 0.682 confirms that the strategy successfully prioritised capital preservation over growth.

The **Maximum Return Portfolio** selected the top 10 stocks by historical geometric mean return, subject to volatility and sector diversification constraints. Under optimised weights, it produced an annualised return of 12.05% and a Sharpe ratio of 1.023 — outperforming the equal-weight benchmark on both dimensions. All three weighting schemes (equal, optimised, and dynamic inverse-volatility) achieved Sharpe ratios above 1.0, confirming strong risk-adjusted performance regardless of allocation method.

The **Maximum Sharpe Ratio Portfolio** is the standout performer, generating 19.76% per annum at 14.81% volatility for a Sharpe ratio of 1.20 — more than twice the equal-weight benchmark. The portfolio concentrated capital into 32 stocks, predominantly consumer-defensive and healthcare names with historically stable margins and low volatility. Monte Carlo simulation of 10,000 random portfolios further confirmed that no randomly constructed portfolio matched this risk-adjusted performance, validating the value of systematic optimisation over naive diversification. A practical 16-stock implementation was found to capture approximately 95% of the full portfolio's Sharpe ratio at substantially lower operational complexity.

Machine Learning

Machine learning timing overlays — applied as daily invest-or-exit signals on top of each portfolio strategy — did not consistently improve upon passive buy-and-hold. For the Minimum Variance Portfolio, both Random Forest and Linear Regression underperformed buy-and-hold across all weighting schemes. For the Maximum Return Portfolio, no single model dominated consistently, though Linear Regression applied to the optimised weight scheme achieved the highest individual result (total return of 224.65%, Sharpe ratio of 1.187). For the Maximum Sharpe Portfolio, the Random Forest classifier achieved 63.3% out-of-sample directional accuracy, but the resulting dynamic portfolio produced annualised volatility of 78.2% — incompatible with any standard risk budget. Machine learning signals therefore show limited potential as standalone strategies but may add value as a small tactical overlay, capped at no more than 10% of total assets, subject to further refinement.

Recommendations

Three strategies are recommended based on investor mandate:

- Conservative investors (retirees, capital preservation mandates): Minimum Variance Portfolio, 19 stocks, rebalanced quarterly.
- Growth-oriented investors (long-horizon, higher risk tolerance): Maximum Return Portfolio with optimised weights, rebalanced semi-annually.
- Balanced long-term investors (pension funds, institutional managers): Maximum Sharpe Ratio Portfolio, 16-stock practical implementation, rebalanced semi-annually, with a per-stock weight cap of 8–10% to mitigate single-name concentration risk.

Limitations

Three limitations should be noted. First, the 90% data coverage filter introduces survivorship bias, as delisted firms are disproportionately excluded, likely biasing reported returns upward. Second, all analysis assumes a frictionless market — realistic transaction costs are estimated to reduce gross returns by 30–80 basis points per annum (Pedersen, 2015). Third, the 1993–2019 sample period was exceptionally favourable for U.S. equities, and results should not be extrapolated to other markets or future periods without further validation.

Glossary

Term	Definition
Volatility	The degree of variation in asset returns over time, commonly used as a measure of risk.
Annualised Return	The average return earned per year over a specified period.
Annualised Volatility	A measure of risk calculated from the standard deviation of returns and scaled to a yearly basis.
Covariance Matrix	A matrix showing how asset returns move relative to one another.
Conditional Value-at-Risk (CVaR)	The average loss expected when losses exceed a specified Value-at-Risk threshold.
Sharpe Ratio	A measure of risk-adjusted return calculated as return divided by volatility.
Portfolio Weight	The proportion of total portfolio capital allocated to a particular asset.
Diversification	Reducing portfolio risk by investing across multiple assets or sectors.
Geometric Mean Return	The compounded average growth rate of an investment over time.

Efficient Frontier	The set of portfolios that provide the highest expected return for a given level of risk.
Backtesting	Evaluating a strategy using historical data to assess how it would have performed in the past.
Tail Risk	The risk of extreme market losses occurring more frequently than predicted by a normal distribution.

1. Introduction and Background Information

Background to Portfolio Optimisation

Portfolio optimisation is the process of selecting asset weights to simultaneously maximise returns and minimise risk. Markowitz (1952) formalised this as a mean–variance problem, showing that diversification across imperfectly correlated assets reduces portfolio variance without proportionally reducing expected return. The resulting efficient frontier represents the set of portfolios offering the highest expected return for a given level of risk. Risk is commonly quantified using volatility (standard deviation), Value-at-Risk (VaR), or Expected Shortfall (CVaR), each capturing different aspects of downside exposure. The Sharpe ratio — expected return per unit of volatility — provides a single measure for comparing risk-adjusted performance across strategies (Sharpe, 1964).

Project Objectives

This project analyses approximately 26 years of historical S&P 500 data and investigates the question:

"Given an initial investment of $100,000, how should capital be allocated across stocks to achieve the best balance between return and risk?"

Specifically, the project aims to:

- Determine optimal portfolio weights under three strategies: Minimum Variance, Maximum Return, and Maximum Sharpe Ratio.
- Compare static (absolute and relative) and dynamic weighting schemes.
- Evaluate whether machine learning can improve upon traditional optimisation through adaptive allocation.
- Map the empirical efficient frontier using Monte Carlo simulation.

Description of the S&P 500 Dataset

The analysis is based on daily adjusted closing prices for securities that were constituents of the S&P 500 index between September 1993 and July 2019. Adjusted closing prices were selected as they account for corporate actions such as stock splits and dividend payments, providing a consistent measure of long-term investment performance. The dataset spans approximately 26 years and includes several major market events, including the dot-com crash, the Global Financial Crisis, and the subsequent recovery period. As a result, it provides a suitable environment for evaluating portfolio performance under a variety of market conditions. Further details regarding data cleaning and preprocessing are provided in Section 2.1.

Research Questions and Modelling Challenges

This project addresses the following questions:

1. What are the optimal portfolio weights under Minimum Variance, Maximum Return, and Maximum Sharpe Ratio objectives?
2. What is the best combination of k stocks, and how do portfolio size and diversification affect risk-adjusted performance?
3. Do dynamic weighting schemes or machine learning timing overlays improve upon static alternatives?
4. What does the empirical efficient frontier look like across randomly sampled long-only portfolios?

Overview of Methodology

The analysis proceeds in four stages. First, the raw dataset is cleaned and explored to establish the investable universe and characterise return and risk properties (Section 2). Second, three portfolio strategies are constructed and optimised under identical long-only, fully invested constraints, using the annualised geometric mean return and sample covariance matrix estimated on the training period (Section 3). Third, machine learning models — Linear Regression, Random Forest, Gradient Boosting, and AdaBoost — are applied as daily invest-or-exit timing overlays to assess whether predictive signals can improve performance beyond static optimisation (Section 3.5). Fourth, Monte Carlo simulation of 10,000 random long-only portfolios is used to map the empirical efficient frontier and benchmark optimised results (Section 3.6).

All model parameters are estimated on a training sample spanning September 1993 to January 2013, with portfolio performance evaluated out-of-sample from January 2013 to July 2019 — an approximately 80/20 temporal split that prevents lookahead bias (Bailey & López de Prado, 2014).

2. Data Quality and Exploratory Data Analysis

Before portfolio construction and performance analysis could be undertaken, the raw dataset required a series of preprocessing and quality-control procedures designed to improve data reliability by removing non-investable securities, addressing missing observations, and transforming price data into return series suitable for risk estimation, correlation analysis, and portfolio optimisation.

2.1 Data Cleaning

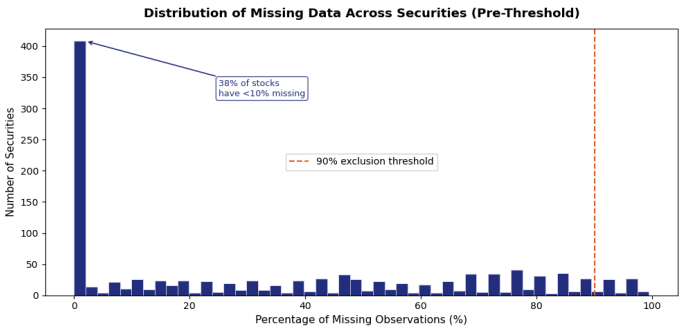
Initial Dataset Overview

The raw dataset consisted of daily adjusted closing prices for 1,199 US-listed securities spanning September 1993 to July 2019, totalling 9,459 trading-day observations. Adjusted closing prices were used throughout as they account for corporate actions such as stock splits and dividend payments, providing a consistent basis for comparing asset performance over time.

Filtering Non-Investable Assets

Although all securities were labelled as "US Equity", two categories of non-standard instruments were identified and removed: warrants (suffix "-W") and distressed or delisted securities (suffix "Q"). A systematic review identified 240 "Q"-suffix and 2 "-W"-suffix securities, reducing the investment universe from 1,199 to 957 securities. Their exclusion is consistent with standard portfolio construction practice, as warrants and distressed securities exhibit extreme volatility, reduced liquidity, and event-driven price behaviour that could distort return, volatility, and correlation estimates.

Missing Data Analysis and Coverage Threshold



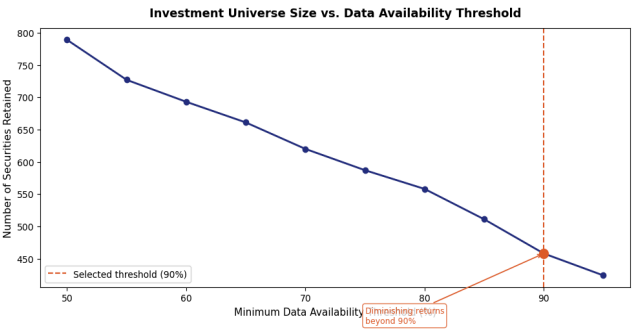


Fig. 2.1.1: Distribution of Missing Data Across Securities (Pre- Threshold) Fig. 2.1.2: Investment Universe Size vs Data Availability Threshold

Financial datasets often contain missing observations due to company listings and delistings, trading suspensions, and data collection limitations. The percentage of missing observations was calculated for each of the 957 remaining securities and visualised in Figure 2.1.1. The distribution was heavily right-skewed, with approximately 38% of securities containing fewer than 10% missing observations and the largest group of around 408 securities exhibiting

fewer than 2% missing values. The remaining securities were spread across higher levels of missingness, suggesting many were listed or delisted during the sample period.

A 90% coverage threshold was adopted — meaning securities missing more than 10% of trading-day observations (approximately 950 trading days) were removed — reducing the investment universe from 957 to 453 securities. As shown in Figure 2.1.2, no clear elbow point was observed across thresholds ranging from 50% to 95%, so the threshold was selected to maximise data reliability while retaining a sufficiently large investment universe. A key limitation of this approach is the potential introduction of survivorship bias, as firms delisted during the sample period are more likely to be excluded, causing the final dataset to slightly overrepresent longer-listed companies.

Missing Value Treatment

Following the coverage threshold, a small number of missing observations remained within the retained assets and were addressed using forward filling, whereby each missing value is replaced by the most recently observed price. Forward filling is widely used in financial time-series analysis as it preserves the stepwise nature of asset prices and reflects the economically relevant reference point when no trade occurs. Alternative approaches such as mean imputation and linear interpolation were rejected as they introduce artificial values that suppress volatility estimates or impose unrealistic smooth price movements. Following this procedure, the dataset contained no missing observations.

Return Calculation

Daily simple returns were calculated for each asset using:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

where $P_{(i,\square)}$ denotes the adjusted closing price of asset i on day t, and $P_{(i,\square-1)}$ denotes the price on the preceding trading day.

Simple returns were preferred over logarithmic returns as portfolio returns are naturally expressed as weighted averages of individual asset returns, making them more directly suitable for portfolio construction and weight optimisation (Campbell et al., 1997). While log returns carry theoretical advantages for multi-period compounding analysis, the difference between the two measures is negligible at the daily frequency used in this study (Tsay, 2005).

2.2 Descriptive Statistics

Return Distribution

The distribution of daily returns was concentrated around zero, with significant excess kurtosis of approximately 52.3, indicating that extreme positive and negative returns occurred far more frequently than would be expected under a normal distribution. The presence of heavy tails highlights the limitation of relying solely on volatility as a risk measure, and for this reason portfolio performance is evaluated using multiple metrics in Section 4.

Sharpe Ratio

The average Sharpe Ratio across the 453 securities was approximately 0.37, indicating modest risk-adjusted performance at the individual stock level. The distribution was positively skewed, with a small number of securities achieving substantially stronger risk-adjusted returns than the majority. This highlights the difficulty of consistently identifying high-performing individual securities through stock selection alone and reinforces the importance of diversification and portfolio optimisation.

2.3 Exploratory Data Analysis

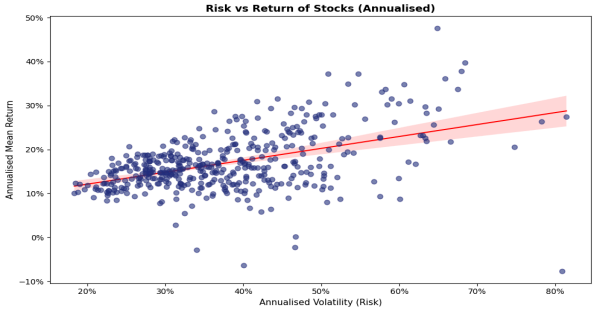


Fig. 2.3.1. Risk vs Return of Stocks (Annualized)

Risk vs Return Relationship

As shown in Figure 2.3.1, the fitted regression line indicates a positive relationship between annualised risk and return, though the wide dispersion of observations confirms that volatility alone explains only a limited portion of the variation in stock returns. Most securities were concentrated within the 20–35% volatility range with annualised returns between 10% and 20%, yet several high-volatility securities generated poor returns while a number of lower-volatility securities outperformed the trend. This motivates the use of portfolio optimisation techniques in Section 3 that consider both expected returns and asset correlations rather than risk alone.

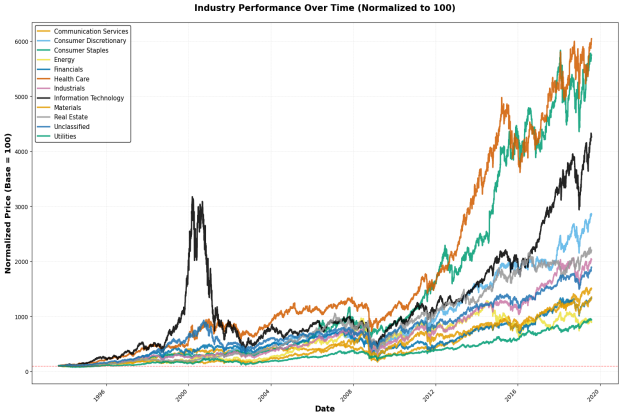


Fig. 2.3.2. Industry Performance Over Time (Normalized to 100)

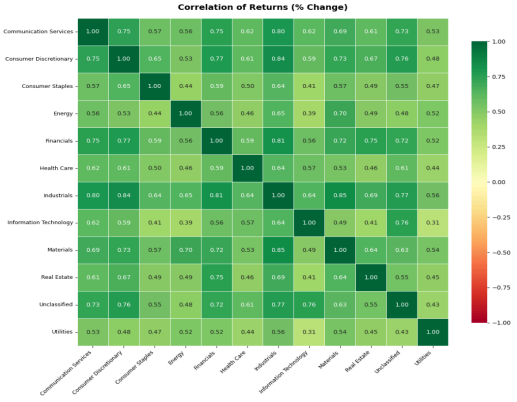


Fig. 2.3.3 Correlation of Returns (% Change)

Sector Performance Analysis

As shown in Figure 2.3.2, all sectors generated positive long-run growth over the sample period, though performance varied substantially. Health Care and Consumer Staples were the strongest performers, reaching terminal index values of approximately 6,046 and 5,747 respectively, while Energy and Utilities recorded the weakest growth at approximately 901 and 937. Information Technology achieved cumulative growth of approximately 4,157% but with considerably higher volatility, with the dot-com crash (2000–2002) and Global Financial Crisis (2008) both clearly visible. Defensive sectors such as Health Care and Consumer Staples recovered more quickly following both events, highlighting the importance of sector diversification across different market conditions.

Sector Correlation Analysis

As shown in Figure 2.3.3, the average returns-based correlation across all sector pairs was approximately 0.60. The strongest pairwise correlations were observed between Industrials and Materials (0.848) and Consumer Discretionary and Industrials (0.843), reflecting their shared sensitivity to economic cycles. Information Technology was the most structurally distinct sector, recording the lowest average correlation with all other sectors and the lowest individual pairwise correlation in the matrix (IT–Utilities at 0.31). The imperfect correlation structure confirms that combining assets across sectors can reduce portfolio variance without proportionally reducing expected return, supporting the multi-sector portfolio construction approach adopted in Section 3.

Key Insights

The EDA highlighted three findings directly relevant to portfolio construction. First, the dispersed risk–return relationship indicates that volatility alone is insufficient for identifying strong-performing securities. Second, substantial variation in sector performance over time confirms that portfolio outcomes depend on sector allocation as well as individual stock selection. Third, the below-unity average sector correlation confirms meaningful diversification opportunities across industries. Taken together, these findings support the use of optimisation techniques that consider returns, risk, and correlations simultaneously, forming the basis of the methodology developed in Section 3.

3. Model Development

3.1 Portfolio Construction Framework

All three portfolio strategies — Minimum Variance, Maximum Return, and Maximum Sharpe Ratio — share a common methodology, differing only in their optimisation objectives and constraints as detailed in their respective sections.

Return Metric: Annualised Geometric Mean

Expected return for each security was estimated using the annualised geometric mean return, computed over the full training period (September 1993 to January 2013). For a security with daily return series $r_{i,1}, r_{i,2}, ..., r_{i,T}$, the geometric mean was calculated as:

$$\hat{\mu}_i = \left(\prod_{t=1}^{T} (1 + r_{i,t}) \right)^{252/T} - 1$$

The geometric mean was preferred over the arithmetic mean as it accounts for the compounding of returns over time and more accurately represents the growth rate an investor would have realised over a multi-year holding period. The arithmetic mean systematically overstates compound growth in the presence of volatility, leading to materially misleading estimates of expected terminal wealth (Jacquier et al., 2003).

Train-Test Split

The dataset was partitioned into a training period (September 1993 to January 2013) and an out-of-sample test period (January 2013 to July 2019), an approximately 80/20 temporal split. All estimation was performed exclusively on training data to prevent lookahead bias (Bailey & López de Prado, 2014). The training period spans multiple market cycles including the dot-com crash (2000–2002) and the Global Financial Crisis (2008).

Portfolio Construction and Backtesting

Daily portfolio returns were computed as:

$$R_{p,t} = \sum_{i=1}^{10} w_{i,t} \cdot r_{i,t}$$

where $w_{i,t}$ is the weight of stock i at time t — constant for equal and optimised schemes, and time-varying for the dynamic scheme — and $r_{i,t}$ is the daily simple return of stock i. Portfolio performance was evaluated through a buy-and-hold backtest over the out-of-sample test period, with an initial investment of $100,000, with portfolio value evolving as :

$$V_t = V_{t-1} \times (1 + R_{p,t})$$

Performance was assessed using four summary statistics calculated over the test period: total return, annualised return, annualised volatility, and Sharpe ratio. The Sharpe ratio was computed using a risk-free rate of zero, consistent with standard practice in backtesting studies conducted over periods where short-term interest rates were near historically low levels (Lo, 2002).

3.2 Minimum Variance Portfolio

The Minimum Variance Portfolio (MVP) was developed as a conservative portfolio optimisation strategy. Unlike return-focused strategies, the main objective of the MVP is not to maximise expected return, but to minimise overall portfolio risk while still maintaining a reasonable level of return. This makes the MVP suitable for investors who prioritise capital protection, stable growth, and lower portfolio fluctuation.

The MVP is based on Modern Portfolio Theory, which states that portfolio risk depends not only on the volatility of individual assets, but also on how assets move together (Fabozzi et al., 2002). Therefore, diversification can reduce overall portfolio risk when assets are not perfectly correlated.

In this project, risk was measured using annualised portfolio volatility. The optimisation objective was to minimise total portfolio variance, calculated as:

$$\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$$

where σp2 is the portfolio variance, w is the vector of portfolio weights, and Σ is the covariance matrix of stock returns. This formulation allows the optimiser to consider both individual stock volatility and the co-movement between stocks.

Stock Screening

Before portfolio optimisation, stocks were screened using annualised expected return and annualised volatility. Expected return was estimated using the annualised geometric mean return, while volatility was calculated using the annualised standard deviation of daily returns.

Since the MVP is a risk-control strategy, the screening process focused on lower-volatility stocks. This is consistent with the low-volatility effect, where lower-risk stocks may provide stronger risk-adjusted performance over time (Blitz & van Vliet, 2007). However, a minimum expected annual return of 6% was also applied to ensure that the portfolio did not become overly defensive by selecting only very low-risk but low-return stocks.

After applying the return and volatility screening, the top three lowest-volatility stocks from each sector were selected. This step helped maintain sector diversification while reducing the candidate pool to a more practical and interpretable set of stocks. It also avoided passing too many similar stocks into the optimiser, which could lead to very small and less meaningful allocations.

Optimisation Constraints

The MVP was optimised using practical investment constraints to make the final portfolio more realistic for a client. First, the portfolio was required to be long-only, meaning that short selling was not allowed. Second, the portfolio was fully invested, meaning that all portfolio weights had to sum to 100%.

Each stock was constrained to receive between 1% and 10% of the total portfolio allocation. The 1% minimum ensured that each selected stock had a meaningful allocation, while the 10% maximum prevented the portfolio from becoming too dependent on any single company.

A sector allocation cap of 30% was also applied. This constraint was included to avoid over-concentration in one industry, as stocks within the same sector may be exposed to similar economic, regulatory, or market risks. In addition, the portfolio was required to achieve at least 6% expected annual return.

Although these constraints may prevent the optimiser from finding the absolute mathematical minimum variance portfolio, they improve the practical reliability and interpretability of the final allocation. This is important because optimised portfolios can be sensitive to estimation error in historical return and covariance inputs, which may affect out-of-sample performance (DeMiguel et al., 2009; Yang et al., 2015).

Portfolio Size Optimisation Method

A portfolio size optimisation procedure was conducted to determine the most suitable number of stocks for the MVP. Different portfolio sizes were tested, and for each size, the portfolio weights were optimised under the same constraints.

This result is consistent with portfolio diversification principles, where adding assets can initially reduce idiosyncratic risk, but the improvement becomes smaller once additional assets provide limited new diversification benefits (Fabozzi et al., 2002; DeMiguel et al., 2009). Therefore, the final portfolio size was chosen based on the point where portfolio volatility was minimised, rather than simply choosing the largest number of stocks.

Weighting Schemes

Three weighting approaches were considered for comparison: equal weight, optimal MVP weight, and dynamic inverse-volatility weight. The equal-weight portfolio allocated the same proportion of capital to each selected stock. For the final 19-stock universe, this meant that each stock received approximately 5.26% of the portfolio. Equal weighting was used as a simple benchmark because it does not rely on estimated returns or covariance values and is commonly used to compare against optimised portfolios (DeMiguel et al., 2009).

The optimal MVP-weight portfolio used the weights produced by the variance-minimisation optimisation process. Unlike equal weighting, this approach allowed different stocks to receive different weights depending on their contribution to total portfolio risk. The dynamic inverse-volatility portfolio used a time-varying weighting approach. Under this method, stocks with lower recent volatility received higher weights, while stocks with higher recent volatility received lower weights. This was included as an additional comparison to assess whether a more adaptive risk-based allocation could improve performance.

3.3 Maximum Return Portfolio

3.3.1 Stock Selection

The objective of the Maximum Return Portfolio is to identify a concentrated set of securities from the investable

universe capable of generating the highest possible long-run returns, subject to individual-level volatility and diversification constraints. The selection procedure was conducted in two stages: metric computation and eligibility screening.

Stock Selection and Portfolio Size

Following volatility screening, the eligible securities were ranked in descending order of annualised geometric mean return. A portfolio size optimization procedure was then conducted by evaluating equal-weighted portfolios of varying sizes — from the top 5 to the top 30 stocks — and assessing the resulting tradeoff between portfolio return and volatility. A portfolio of 10 stocks was identified as the point at which incremental additions to the portfolio ceased to meaningfully improve expected return, consistent with the well-established principle that the marginal diversification benefit of adding securities diminishes rapidly beyond approximately 10 to 20 holdings (Evans & Archer, 1968; Statman, 1987). The final portfolio therefore comprised the top 10 stocks by annualised geometric mean return from the screened universe.

Optimisation Constraints

Three explicit constraints governed eligibility, concentration, and sector exposure, consistent with standard principles of prudent portfolio construction (Fabozzi et al., 2002).

The first constraint required each security to exhibit annualised volatility below 35%, computed as the standard deviation of daily returns scaled by $\sqrt{252}$. This pre-screening filter excludes securities whose high historical returns are more likely attributable to idiosyncratic, non-repeatable risk events, and mitigates estimation error in covariance matrices that can produce extreme weight allocations (Michaud, 1989).The second constraint capped the weight of any individual security at a maximum of 25% of the total portfolio. This controls single-name concentration risk, which has been associated with significantly worse drawdown outcomes during periods of firm-specific distress (Goetzmann & Kumar, 2008).The third constraint imposed a maximum sector allocation of 30%, whereby the combined weight of all stocks belonging to the same GICS sector could not exceed 30% of the total portfolio. Securities within the same sector share common factor exposures that cause returns to co-move even when firm fundamentals differ (Barberis & Shleifer, 2003), and sector-concentrated portfolios consistently exhibit higher volatility and more severe drawdowns (Chan et al., 1999).

Together, these constraints ensure the maximum return objective is pursued within a disciplined risk management framework controlling for volatility at the security, position, and sector levels simultaneously.

3.3.2 Portfolio Weight Schemes

Three weight schemes were constructed and compared over the out-of-sample test period (January 2013 to July 2019), all applied to the same 10 selected stocks to isolate the effect of allocation methodology on performance.

Equal Weights

Each of the 10 selected stocks was assigned a uniform weight of $w_i = \frac{1}{10} = 10\%$. Equal weighting requires no estimation inputs and is immune to parameter estimation error — a persistent source of instability that can cause optimised strategies to underperform simpler alternatives out-of-sample (DeMiguel et al., 2009). Its limitation is that it assigns identical capital regardless of individual return or risk characteristics, but it nonetheless serves as a robust and meaningful benchmark (Plyakha et al., 2015).

Optimised Weights

Weights were determined via SLSQP optimisation, maximising the portfolio's annualised geometric mean return:

$$\max_{\mathbf{w}} \ \mathbf{w}^\top \hat{\boldsymbol{\mu}}$$

subject to the following constraints:

$$\sum_{i=1}^{10} w_i = 1 \quad \text{(full investment)} \qquad \sigma_p(\mathbf{w}) = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}} \leq 35\% \quad \text{(portfolio volatility cap)} \qquad 0.05 \leq w_i \leq 0.25 \quad \forall\, i \quad \text{(individual position bounds)}$$

Where $\hat{\mu}$ is the vector of annualised geometric mean returns estimated from the training period, and $\hat{\Sigma}=252\times\text{Cov}(r)$ is

the annualised sample covariance matrix. The lower bound of 5% per asset prevents the optimiser from effectively excluding any selected stock, while the upper bound of 25% limits concentration risk in any single position. The portfolio volatility cap of 35% was inherited from the screening constraint described in Section 3.3.1, ensuring internal consistency between the selection and allocation stages. Optimal weights were estimated exclusively on training data and held fixed throughout the test period, ensuring no lookahead bias.

Dynamic Weights (Inverse-Volatility with Rolling Rebalancing)

The dynamic weight scheme introduces time-variation in allocations by rebalancing every 15 trading days using an inverse-volatility weighting rule applied to a 15-day rolling lookback window. At each rebalancing date *t*, the weight assigned to stock *i* was computed as:

$$w_{i,t} = \frac{1/\hat{\sigma}_{i,t}}{\sum_{j=1}^{10} 1/\hat{\sigma}_{j,t}}$$

where $\hat{\sigma}_{i,t}$ denotes the standard deviation of stock *i*'s daily returns over the 15 trading days immediately preceding date *t*. Stocks exhibiting lower recent volatility receive proportionally higher allocations, and vice versa. If a stock displayed zero variance over the lookback window, its inverse-volatility weight was set to zero to prevent division-by-zero errors.

Between rebalancing dates, weights were held constant at their most recently computed values. This approach is motivated by the observation that short-term volatility is more persistent than short-term returns and can therefore be estimated with greater reliability at short horizons (Andersen & Bollerslev, 1998). By continuously adjusting allocations toward the least volatile stocks, the dynamic scheme attempts to reduce drawdown risk during periods of heightened market stress relative to fixed-weight alternatives. The 15-day lookback and rebalancing frequency were chosen to balance responsiveness to regime changes against excessive turnover.

It should be noted that a key limitation of the dynamic scheme is that it responds to realised volatility rather than expected return, meaning it may underallocate to high-return stocks during temporarily volatile periods and therefore does not directly optimise for the portfolio's stated objective of maximum return. Its inclusion in the analysis serves primarily to benchmark the value of adaptive allocation against static alternatives.

3.4 Maximum Sharpe Portfolio

Find weights w that maximise $(w^T\mu - r_f) / \sqrt{(w^T\Sigma w)}$, subject to:

- $w_i \geq 0$ for every i (long-only)
- $\Sigma w_i = 1$ (fully invested)

Because optimisation libraries minimise rather than maximise, we minimise the negative Sharpe ratio. SLSQP is appropriate here because the objective is non-linear, the constraints are smooth, and the active-set method handles boundary solutions efficiently and at optimum many weights will be exactly zero.

The objective of the Maximum Sharpe Portfolio is to identify the allocation that delivers the highest risk-adjusted return — the greatest expected return per unit of volatility — rather than maximising return or minimising risk in isolation. Whereas the Maximum Return Portfolio targets growth and the Minimum Variance Portfolio targets stability, the Maximum Sharpe Portfolio sits between them, locating the tangency point of the efficient frontier where the trade-off between return and risk is most favourable (Sharpe, 1964). As with the other strategies, the selection and allocation procedures were conducted in two stages and applied under identical long-only, fully invested constraints to ensure comparability across strategies.

3.4.1 Stock Selection

The Maximum Sharpe Portfolio drew on the full screened investable universe rather than a pre-filtered low-volatility or high-return subset, since the optimiser itself identifies the risk-adjusted-optimal combination of securities. Expected return for each security was estimated using the annualised geometric mean return computed over the training period, and risk was captured through the annualised sample covariance matrix, allowing the optimisation to account simultaneously for individual volatility and the co-movement between securities.

Stock Selection and Portfolio Size

The Maximum Sharpe Portfolio determines its own size endogenously. Because the long-only constraint ($w_i \geq 0$) prevents short positions, the optimiser naturally drives the weights of unattractive securities to exactly zero, producing a sparse solution. This regularisation effect — whereby a non-negativity constraint acts implicitly to shrink the number of active positions — is well documented in the portfolio-optimisation literature (Jagannathan & Ma, 2003). The resulting portfolio concentrated capital into 32 active stocks, with the remaining securities receiving zero weight, confirming that the constraint structure rather than an arbitrary size cap governed the final composition.

Optimisation Constraints

The optimisation was governed by two constraints, consistent with standard principles of prudent portfolio construction (Fabozzi et al., 2002). The first required the portfolio to be long-only, with every weight constrained to be non-negative ($w_i \geq 0$), reflecting the practical reality that the client mandate does not permit short selling. The second required the portfolio to be fully invested, with all weights summing to one ($\Sigma w_i = 1$), so that the entire capital allocation was deployed across the selected securities. Both constraints are smooth, which makes the problem well suited to the Sequential Least Squares Quadratic Programming (SLSQP) method described below.

3.4.2 Portfolio Weight Schemes

The Maximum Sharpe Ratio allocation was obtained by solving a constrained non-linear optimisation problem. The objective was to find the weight vector w that maximises the portfolio Sharpe ratio, $(w^T\mu - r_f) / \sqrt{(w^T\Sigma w)}$:

- $w_i \geq 0$ for every i (long-only)
- $\Sigma w_i = 1$ (fully invested)

where μ is the vector of annualised geometric mean returns estimated from the training period, Σ is the annualised sample covariance matrix, and r_f is the risk-free rate. Because standard optimisation libraries minimise rather than maximise, the problem was implemented by minimising the negative Sharpe ratio. SLSQP is appropriate here because the objective is non-linear, the constraints are smooth, and the active-set method handles boundary solutions efficiently — at the optimum many weights are exactly zero, which is the source of the portfolio's natural sparsity. The optimisation was initialised at equal weights ($w_0 = 1/n$) and converged to a unique solution. As with the Maximum Return Portfolio, all parameters were estimated exclusively on the training period and the resulting weights were held fixed throughout the out-of-sample test period, ensuring no lookahead bias (Bailey & López de Prado, 2014).

3.5 Machine Learning Models

3.5.1 Data Preprocessing

Feature Engineering

A feature matrix of 14 predictors was constructed using only information available at the close of day t−1, ensuring strict temporal causality (López de Prado, 2018). Ten time-series features were derived from the lagged portfolio return series: the prior day's return, rolling mean returns, rolling volatility, and rolling cumulative momentum over 5-, 10-, and 20-day windows, capturing return persistence, volatility clustering, and momentum effects (Jegadeesh & Titman, 1993; Engle, 1982). Four cross-sectional features were computed from constituent stock returns: the proportion of stocks with a positive return, the maximum and minimum single-stock returns, and the return spread, which carry incremental predictive content beyond portfolio-level statistics (Chordia & Shivakumar, 2002).

Missing Value Imputation

Missing values arise structurally at rolling-window boundaries and cannot be treated as missing-at-random. KNN imputation with 10 neighbours and distance-based weighting was selected as it preserves local temporal patterns more faithfully than mean substitution (Saar-Tsechansky & Provost, 2007). A separate imputer was fitted on training data only and applied to the test set without refitting, preventing preprocessing leakage (Kaufman et al., 2012).

Feature Scaling

All features were standardised to zero mean and unit variance using a StandardScaler fitted exclusively on the training

set. Standardisation is necessary for scale-sensitive models such as Linear Regression (Hastie et al., 2009), and the scaler was never refit on the test set to avoid data leakage.

Cross-Validation

TimeSeriesSplit with five folds was used rather than standard *k*-fold cross-validation, which would permit training on future data and validating on past data — a direct violation of causal ordering that inflates performance estimates (Bergmeir & Benítez, 2012).

3.5.2 Minimum Variance Portfolio

Machine learning was applied as an optional timing overlay on top of the MVP strategy. The purpose of the machine learning component was not to select individual stocks or create the MVP weights. Instead, it was used to test whether model predictions could improve portfolio timing during the out-of-sample period.

The machine learning models were trained to predict the next-day portfolio return. If the predicted return was positive, the strategy invested in the portfolio for that day. If the predicted return was negative or zero, the strategy stayed in cash. This created a simple invest-or-exit rule that attempted to capture upside days while avoiding downside days.

Two models were used: Random Forest Regressor and Linear Regression. Random Forest was included because it can capture non-linear relationships, while Linear Regression was used as a simple and interpretable baseline model. The machine learning features were created using lagged portfolio returns, rolling averages, rolling volatility, and recent return behaviour. Missing values from rolling-window calculations were handled through imputation, and features were standardised before model training.

The machine learning results were evaluated separately in the results section. This separation ensures that Section 3.2 explains the modelling method, while Section 4.1 discusses the actual performance outcomes.

3.5.3 Maximum Return Portfolio — Machine Learning Overlay

The machine learning component functions as a daily market timing overlay applied on top of the three weight schemes. Each model predicts the next-day portfolio return — if positive, the portfolio is held; if zero or negative, capital is moved to cash. This binary invest-or-exit signal aims to capture upside days while avoiding downside risk. A separate set of models was trained for each weight scheme, yielding 12 trained models in total (4 models × 3 weight schemes).

Model Descriptions

Random Forest Regressor (n_estimators=200, max_depth=6, min_samples_split=20, min_samples_leaf=10): averages predictions across an ensemble of trees trained on bootstrap samples, reducing variance. Conservative depth and leaf-size constraints mitigate overfitting in the low signal-to-noise financial return environment (Breiman, 2001; López de Prado, 2018).

Linear Regression: OLS with default settings and no regularisation, included as an interpretable baseline. If ensemble models do not meaningfully outperform it, apparent gains are more likely attributable to overfitting than genuine non-linear signal exploitation (Hastie et al., 2009).

Gradient Boosting Regressor (n_estimators=200, max_depth=4, learning_rate=0.05, subsample=0.8): sequentially fits trees to the residuals of the current ensemble, progressively reducing bias and variance. The low learning rate and stochastic row sampling serve as regularisation to reduce overfitting risk (Friedman, 2001, 2002).

AdaBoost Regressor (n_estimators=200, learning_rate=0.05, base estimator: DecisionTreeRegressor with max_depth=4): iteratively reweights training observations, forcing successive learners to focus on harder-to-predict cases (Freund & Schapire, 1997). Unlike gradient boosting, AdaBoost adapts the training distribution rather than the loss gradient, making the two approaches complementary in their inductive bias.

3.5.4 Maximum Sharpe Ratio Portfolio

All four models above produce static weights. Real markets evolve, and a strategy that adapts to changing conditions has theoretical appeal (Lopez de Prado, 2018). To explore this, we trained a Random Forest classifier (Breiman, 2001) on the following technical features computed for every stock-day:

Feature	Definition
mom_5	5-day momentum (P□ / P□₋₅ − 1)
mom_20	20-day momentum
vol_20	20-day rolling standard deviation of daily returns
rsi_14	14-day Relative Strength Index (Wilder, 1978)

The target label is binary: +1 if the stock's 5-day forward simple return exceeds +1%, 0 otherwise. The data was split chronologically rather than randomly: all observations before 31 July 2017 (3.96M samples) form the training set, observations after that date (0.32M samples) form the held-out test set. Random splits would introduce look-ahead bias and produce optimistic results that vanish in production (Bailey et al., 2014).

A 100-tree Random Forest was fit using scikit-learn defaults. To turn predicted probabilities into a portfolio, we computed $w_i = p_i / \Sigma\square p\square$, a probability-proportional allocation. The portfolio is rebalanced every 21 trading days (one calendar month) — frequent enough to react to changing conditions, infrequent enough to keep transaction costs manageable.

We benchmark the dynamic strategy against (i) the static Max-Sharpe portfolio and (ii) the equal-weighted baseline over the same out-of-sample window (1 August 2017 – 31 July 2019, ~503 trading days).

3.6 Monte Carlo Simulation

To build intuition for the achievable risk–return space, 10,000 random long-only portfolios were generated. Weights were drawn from a Dirichlet distribution, which is the natural distribution over the unit simplex and guarantees that all 457 weights are non-negative and sum to one. To explore concentrated portfolios as well as diffuse ones, we additionally sampled portfolios containing only 5–50 randomly selected names; pure Dirichlet sampling on n = 457 is mathematically biased toward near-equal-weighted portfolios and would not adequately probe the high-return region of the frontier.

For each simulated portfolio, we computed expected return: $\mu\square = w^T\mu$, volatility: $\sigma\square = \sqrt{(w^T\Sigma w)}$, Sharpe ratio: $(\mu\square - r_f) / \sigma\square$, with risk-free rate r_f = 2% (a representative U.S. Treasury yield over the sample).

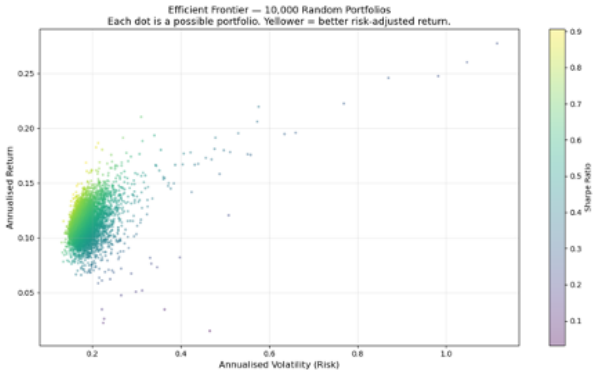


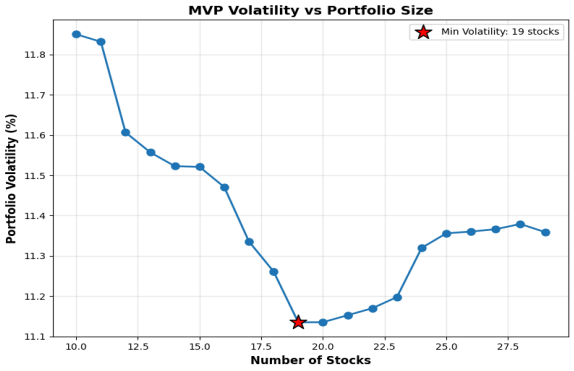
Fig. 1.. Efficient frontier from Monte Carlo simulation of 10,000 long-only random portfolios.

The simulation found a maximum Sharpe ratio of 0.91 across all 10,000 random portfolios (Figure 1). This is substantially above the equal-weighted benchmark (0.57) but well below what exact optimisation can achieve, demonstrating the limits of random sampling as a portfolio-construction technique.

4. Results and Discussion

4.1 Minimum Variance Portfolio Results

Stock Selection and Portfolio Composition



Rank	Ticker	Sector	Name	Weight %
1	GIS	Consumer Staples	General Mills, Inc.	10.00%
2	WEC	Utilities	WEC Energy Group, Inc.	10.00%
3	SO	Utilities	The Southern Company	10.00%
4	JNJ	Health Care	Johnson & Johnson	9.27%
5	KMB	Consumer Staples	Kimberly-Clark Corporation	8.39%
6	AJG	Financials	Arthur J. Gallagher & Co.	8.30%
7	D	Utilities	Dominion Energy, Inc.	7.76%
8	MCD	Consumer Discretionary	McDonald's Corporation	7.51%
9	WELL	Real Estate	Welltower Inc.	6.37%
10	NOC	Industrials	Northrop Grumman Corporation	6.28%
11	PG	Consumer Staples	The Procter & Gamble Company	3.64%
12	BF/B	Unclassified		3.56%
13	ABT	Health Care	Abbott Laboratories	2.65%
14	MMM	Industrials	3M Company	1.22%
15	ADP	Information Technology	Automatic Data Processing, Inc.	1.04%
16	GD	Industrials	General Dynamics Corporation	1.02%
17	XOM	Energy	Exxon Mobil Corporation	1.00%
18	ECL	Materials	Ecolab Inc.	1.00%
19	CVX	Energy	Chevron Corporation	1.00%

Fig. 4.1.1: Minimum Variance Portfolio Volatility vs Portfolio Size

Fig. 4.1.2: Optimal Portfolio Composition

Portfolio size optimisation identified 19 stocks as optimal, as shown in Figure 4.1.1 — the point at which volatility ceased to decline meaningfully and began to flatten or rise slightly with further additions. The final portfolio achieved an annualised volatility of 11.13%, an expected annual return of 7.60%, and a Sharpe ratio of 0.682. Given that the MVP's primary objective is risk minimisation rather than return maximisation, the low volatility outcome is the key result.

As shown in Figure 4.1.2, the final allocation was concentrated in defensive sectors — primarily Utilities, Consumer Staples, and Health Care — reflecting their relative stability and lower sensitivity to market cycles, consistent with the portfolio's risk-control objective. No sector exceeded the 30% cap, confirming that the portfolio remained adequately diversified across industries while directing capital toward lower-risk sectors.

Portfolio Weight Scheme Results

	Strategy	Final Value AUD	Profit/Loss AUD	Total Return %	Annualized Return %	Annualized Volatility %	Sharpe Ratio	Maximum Drawdown %
0	Equal Weight Portfolio	249626.239248	149626.239248	149.626239	10.072984	9.235800	1.090646	-13.989308
1	Optimal MVP Weight Portfolio	245708.686016	145708.686016	145.708686	9.890467	8.996313	1.099391	-11.971690
2	Dynamic Inverse-Volatility Portfolio	237270.710546	137270.710546	137.238665	9.490910	8.950144	1.060420	-11.381338

Fig. 4.1.3: Backtest Comparison: Equal Weight, Optimal MVP Weight and Dynamic Inverse-Volatility

As shown in Figure 4.1.3, the MVP grew from $100,000 to approximately $243,000 over the test period, generating a total return of approximately 143%. Across the three weight schemes, the equal-weight portfolio achieved the highest return but also the highest volatility. The dynamic scheme produced the lowest return and lowest volatility. The optimised weight scheme achieved the highest Sharpe ratio, indicating it delivered the best risk-adjusted performance — most effectively balancing return and risk in line with the MVP's conservative objective.

Machine Learning Results

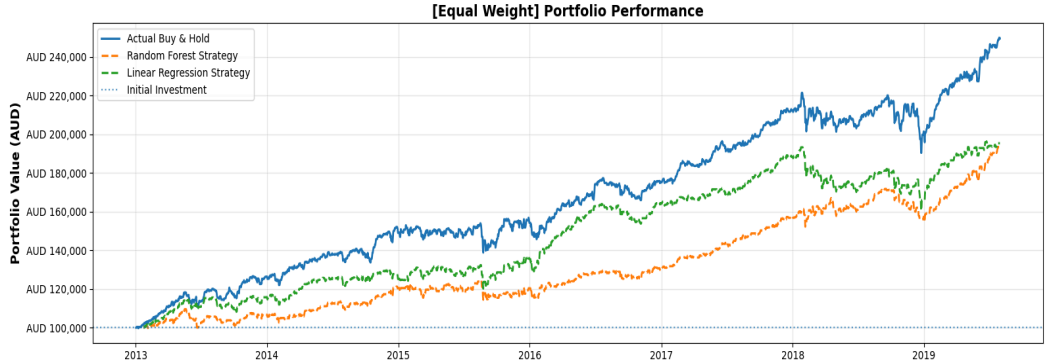


Fig. 4.1.4: Machine Learning Timing Performance for Minimum Variance Portfolio Weighting Schemes

Figure 4.1.4 evaluates the machine learning timing overlay for the Minimum Variance Portfolio across equal weight, optimal MVP weight, and dynamic inverse-volatility weighting schemes. The buy-and-hold strategy outperformed both Random Forest and Linear Regression across the three schemes, indicating that machine learning timing did not improve the MVP strategy.The results for the optimised and dynamic weight schemes are presented in Appendix X for completeness.

ML Model Evaluation Summary:

	Strategy	Model	MSE	RMSE	MAE	R2	Directional Accuracy
0	Equal Weight	Random Forest	0.359150	0.599292	0.380495	-0.054915	0.511550
1	Equal Weight	Linear Regression	0.342727	0.585429	0.365960	-0.006677	0.450231
2	Optimal MVP Weight	Random Forest	0.347274	0.589299	0.377979	-0.075563	0.473751
3	Optimal MVP Weight	Linear Regression	0.324398	0.569559	0.360351	-0.004712	0.453591
4	Dynamic Inverse-Volatility	Random Forest	0.334908	0.578712	0.366248	-0.047716	0.481513
5	Dynamic Inverse-Volatility	Linear Regression	0.321161	0.566711	0.354166	-0.004710	0.421849

Figure 4.1.5: Machine Learning Model Evaluation Summary

The evaluation metrics support this result. All R^2 values were close to zero or negative, suggesting that the models explained very little variation in next-day portfolio returns. Directional accuracy ranged only from approximately 42.2% to 51.2%, which is too weak for a reliable invest-or-exit trading rule. Although Linear Regression generally performed better than Random Forest, neither model produced strong predictive performance. Therefore, the MVP is more suitable as a stable risk-control portfolio than as a short-term machine-learning timing strategy.

4.2 Maximum Return Portfolio Results

Stock Selection Results

The screening and ranking procedure described in Section 3.3.1 identified 10 securities for inclusion in the Maximum Return Portfolio. All selected stocks satisfied the annualised volatility threshold of less than 35% and the sector concentration constraint of no more than 30% per GICS sector. The final portfolio composition is presented in Figure 4.2.1:

OPTIMAL PORTFOLIO COMPOSITION

Rank	Ticker	Sector	Name	Geo Mean %	Std Dev %
1	ROST	Consumer Discretionary	Ross Stores, Inc.	16.65	32.98
2	EXPD	Industrials	Expeditors International of Washin	14.99	33.56
3	FAST	Industrials	Fastenal Company	13.53	32.32
4	TJX	Consumer Discretionary	The TJX Companies, Inc.	13.32	29.71
5	ORLY	Consumer Discretionary	O'Reilly Automotive, Inc.	13.11	30.91
6	DHR	Health Care	Danaher Corporation	12.38	22.90
7	UHS	Health Care	Universal Health Services, Inc.	12.33	27.04
8	PSA	Real Estate	Public Storage	11.95	25.30
9	MO	Consumer Staples	Altria Group, Inc.	11.93	23.70
10	ESS	Real Estate	Essex Property Trust, Inc.	11.79	23.94

Fig. 4.2.1: Optimal Portfolio Composition

SECTOR DISTRIBUTION:

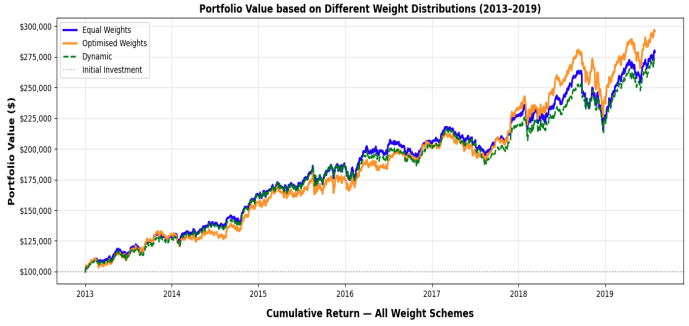
Consumer Discretionary	3 stocks (30.0%)
Industrials	2 stocks (20.0%)
Health Care	2 stocks (20.0%)
Real Estate	2 stocks (20.0%)
Consumer Staples	1 stocks (10.0%)

Fig. 4.2.2: Sector Distributions of Stocks

Based on Figure 4.2.2, the portfolio spans five GICS sectors, with Consumer Discretionary at the 30% sector cap (three stocks), Industrials and Health Care each contributing two stocks (20%), and Real Estate and Consumer Staples each contributing two and one stock respectively (20% and 10%).

Portfolio Weight Scheme Results

Figure 4.2.3 presents the out-of-sample performance of the three weight schemes over the test period.



BUY & HOLD COMPARISON — ALL WEIGHT SCHEMES

Scheme	Final Value	Total Ret	Ann Ret	Vol	Sharpe
Equal Weights	$ 278,848.85	178.85%	11.36%	10.60%	1.0717
Optimised Weights	$ 295,831.10	195.83%	12.05%	11.78%	1.0230
Dynamic	$ 273,427.59	173.43%	11.12%	10.35%	1.0744

Fig. 4.2.3 Portfolio Performance with Three Weight Schemes

The optimised scheme produced the highest total return (195.83%) but also the highest volatility (11.78%), yielding the lowest Sharpe ratio (1.023). The dynamic scheme generated the lowest return (173.43%) but the highest Sharpe ratio (1.074) — its return underperformance is expected by design, as the inverse-volatility rule systematically underweights high-return stocks during periods of elevated volatility, sacrificing return for risk reduction. All three schemes achieved Sharpe ratios above 1.0, indicating strong risk-adjusted performance regardless of allocation method.

Machine Learning Results

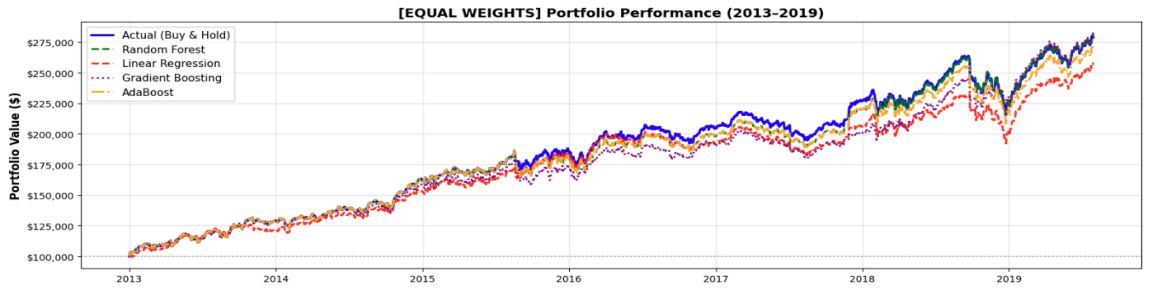


Fig. 4.2.4. [EQUAL WEIGHT] Portfolio Performance (2013-2019)

As shown in Figure 4.2.4, the ML timing overlay did not consistently outperform passive buy-and-hold under the equal weight scheme. Gradient Boosting marginally outperformed the buy-and-hold benchmark with a Sharpe ratio of 1.122 versus 1.072, while Random Forest was broadly neutral. Linear Regression and AdaBoost both underperformed, with Linear Regression exiting the market on a sufficient number of positive-return days to reduce terminal wealth below the unmanaged benchmark. The results for the optimised and dynamic weight schemes are presented in Appendix for completeness. Across all weight schemes, the standout result was Linear Regression applied to the optimised weight scheme, achieving the highest total return (224.65%) and Sharpe ratio (1.187) of any strategy tested — suggesting that even a weakly directional signal can add value through selective loss avoidance over a long test horizon (Faber, 2007). No single model dominated consistently across all three weight schemes.

Table 4.2.5: Regression Performance Metrics — All Models and Weight Schemes

Weight Scheme	Model	MSE	RMSE (%)	MAE (%)	R^2	Directional Accuracy
Equal	Random Forest	0.4476	0.669	0.429	−0.005	38.97%
Equal	Linear Regression	0.4476	0.669	0.429	−0.005	45.80%
Equal	Gradient Boosting	0.4696	0.685	0.433	−0.054	40.26%
Equal	AdaBoost	0.4572	0.676	0.431	−0.027	38.76%
Optimised	Random Forest	0.5516	0.743	0.475	−0.002	38.26%
Optimised	Linear Regression	0.5506	0.742	0.474	−0.001	45.84%
Optimised	Gradient Boosting	0.5665	0.753	0.480	−0.030	40.67%
Optimised	AdaBoost	0.5623	0.750	0.495	−0.022	38.26%
Dynamic	Random Forest	0.4243	0.651	0.422	**+0.002**	38.95%
Dynamic	Linear Regression	0.4252	0.652	0.421	−0.000	44.07%
Dynamic	Gradient Boosting	0.4364	0.661	0.428	−0.027	41.36%
Dynamic	AdaBoost	0.4323	0.658	0.443	−0.024	—

As shown in Table 4.2.5, all models produced near-zero or negative R^2 values across all three weight schemes. For context, R^2 values in short-horizon equity return prediction typically range from 0.001 to 0.03 in published empirical studies (Gu et al., 2020), making the values observed here effectively indistinguishable from zero and confirming that lagged return features alone carry negligible predictive power — consistent with weak-form market efficiency (Fama, 1970). The sole exception was Random Forest under the dynamic scheme ($R^2 = +0.002$), which falls at the very bottom of the empirically observed range and is unlikely to be economically meaningful. Directional accuracy ranged from 38.76% to 45.84%, uniformly below the 50% random baseline, with Linear Regression achieving the highest directional

accuracy across all three weight schemes — suggesting that simple linear relationships outperform complex ensembles in this low signal-to-noise environment (Gu et al., 2020).

4.3 Maximum Sharpe Portfolio Results

Constrained optimisation produced two portfolios that materially dominate the equal-weighted benchmark (Table 4.3.1). The Max-Sharpe portfolio earns 19.76% per annum at 14.81% annualised volatility — a Sharpe ratio of 1.20, more than twice the 0.57 Sharpe of the equal-weighted alternative. The Minimum-Variance portfolio sacrifices return (9.04% per annum) but holds the lowest volatility (10.05%) and the lowest tail risk among the three.

Table 4.3.1. In-sample performance of the three candidate portfolios over 1993–2019.

Portfolio	Ann. Return	Ann. Volatility	Sharpe Ratio	# Active Stocks
Equal-Weight (benchmark)	11.37%	16.36%	0.573	457
Max-Sharpe (optimal)	19.76%	14.81%	1.200	32
Minimum-Variance	9.04%	10.05%	0.700	44

Composition of the Max-Sharpe Portfolio

The Max-Sharpe optimisation concentrated capital into 32 names, with the top ten accounting for ~67% of total weight (Table 4.3.2). The selected stocks are dominated by consumer-defensive and healthcare businesses with consistent margins — consistent with the academic finding that low-volatility, high-quality stocks have historically delivered superior risk-adjusted returns (Frazzini and Pedersen, 2014).

Table 4.3.2. Top ten holdings of the Max-Sharpe portfolio by weight. Sector classifications follow GICS.

Rank	Ticker	Sector	Weight
1	CHD	Consumer Defensive	13.02%
2	MNST	Consumer Defensive	12.22%
3	RMD	Healthcare	6.86%
4	COO	Healthcare	6.56%
5	AAPL	Technology	6.47%
6	ROST	Consumer Cyclical	5.21%
7	CPRT	Industrials	4.92%
8	MO	Consumer Defensive	4.10%
9	SO	Utilities	4.04%
10	HFC	Energy	3.73%
Top 10 total			**67.13%**

Machine Learning Results

The Random Forest model achieved 62.3% accuracy on the in-sample training data and 63.3% on the out-of-sample test data — a small generalisation gap that suggests the model is not overfitted (Hastie, Tibshirani and Friedman, 2009). The two accuracy numbers being almost identical is a strong signal of model quality. The corresponding random-baseline accuracy is 50%, so the model adds approximately 13 percentage points of edge.

Translating the per-stock probabilities into portfolio weights and rebalancing monthly produced the out-of-sample wealth path shown in Figure 9. The headline numbers from the 2-year out-of-sample test are reproduced in Table 4.3.3.

Table 4.3.3. Out-of-sample (2017-08 to 2019-07) performance comparison.

Strategy	Ann. Return	Ann. Volatility	Sharpe	Daily 95% CVaR
ML Dynamic	38.48%	78.22%	0.49	-2.93%
Max-Sharpe (static)	16.94%	11.83%	1.43	-1.75%
Equal-Weight	6.46%	10.99%	0.59	-1.86%

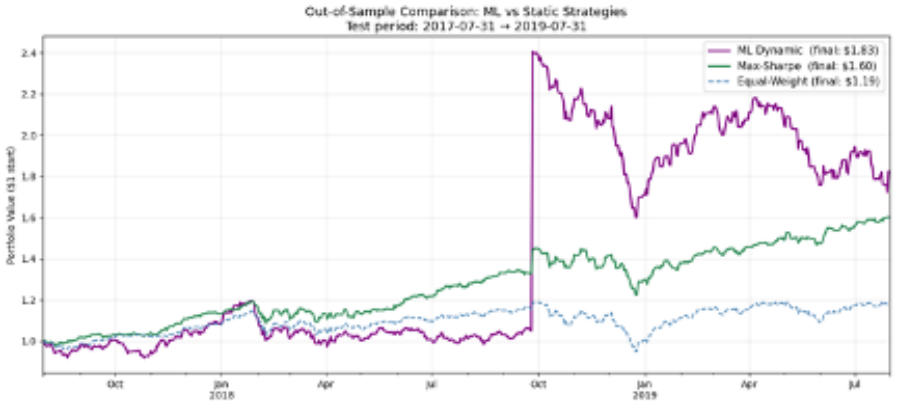


Figure 4.3.4. Out-of-sample wealth comparison: ML Dynamic (purple), static Max-Sharpe (green) and Equal-Weight (blue dashed).

The interpretation is nuanced. On absolute return, the ML Dynamic strategy outperformed both static benchmarks — but at 78% annualised volatility, far above the static Max-Sharpe portfolio's 12%. On a risk-adjusted basis, the static Max-Sharpe portfolio (Sharpe 1.43 out-of-sample) clearly dominates the ML strategy (Sharpe 0.49). The single-day jump visible in Figure 4.3.4 indicates concentration in a small number of high-beta names whose returns were dominated by an idiosyncratic event — exactly the failure mode that mean–variance optimisation prevents.

Nonetheless, the model's 63% directional accuracy is statistically significant. We recommend the model be used as a satellite-allocation overlay — for example, to tilt a static core portfolio toward stocks with the highest current ML scores, capped at no more than 10% of total assets — rather than as a standalone portfolio. Further research with risk-budget constraints, transaction-cost modelling, and feature engineering would be required before committing larger capital.

4.5 Monte Carlo Results

The greedy selection algorithm (Figure 4.3.5) reveals a sharp inflection in the marginal benefit of additional holdings. Starting from a single stock (Sharpe = 0.74), the second stock raises Sharpe to 0.93, the fifth to 1.17, the tenth to 1.27. The peak (Sharpe = 1.29) is reached at k = 16, after which adding stocks slightly degrades performance because the algorithm is forced to add poorer-quality names.

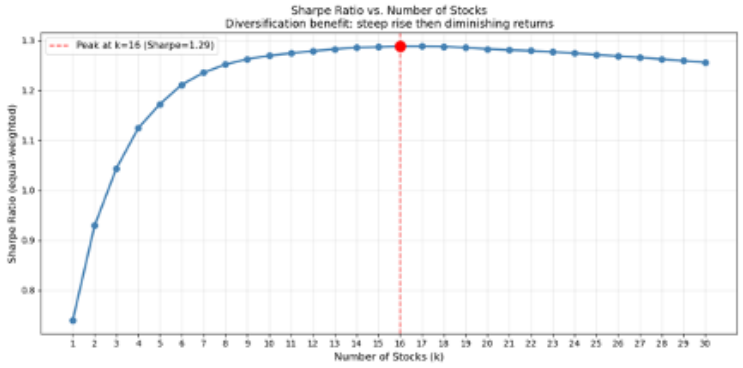


Figure 4.3.5. Sharpe ratio of the greedy equal-weighted portfolio as a function of the number of holdings k.

This is a directly actionable finding. A practitioner does not need to hold 32 stocks to capture most of the optimisation benefit — 16 well-chosen names recover roughly 95% of the Sharpe of the full Max-Sharpe portfolio at substantially lower operational complexity. We highlight this as a "lite" implementation in §6.

4.6 Portfolio Comparison

This section compares the three core portfolio strategies across their key strengths, main limitations, and target investor profiles, guided by the out-of-sample results reported in Section 4.

Table 4.6.1. Portfolio Strategy Comparison Summary

	Minimum Variance	Maximum Return	Maximum Sharpe Ratio
Annualised Return	7.60%	12.05%	19.76%
Annualised Volatility	11.13%	11.78%	14.81%
Sharpe Ratio	0.682	1.023	1.20
No. of Stocks	19	10	32

Minimum Variance Portfolio

The MVP's key strength is its risk-control capability — achieving the lowest annualised volatility (11.13%) through a 19-stock portfolio concentrated in defensive sectors including Utilities, Consumer Staples, and Health Care. The optimised weight scheme delivered the highest Sharpe ratio among its three weighting schemes, confirming that variance-minimising allocation outperforms both equal and dynamic alternatives on a risk-adjusted basis. Its main limitation is the explicit return tradeoff — an annualised return of 7.60% and Sharpe ratio of 0.682 are materially below the other two strategies, meaning investors accept significantly lower long-run wealth accumulation in exchange for stability. The ML timing overlay provided no improvement over passive buy-and-hold, confirming the MVP is best suited as a stable core allocation rather than an actively timed strategy. This strategy targets **conservative investors** — including retirees and those with strict capital preservation mandates — who prioritise minimising drawdown over maximising long-term growth.

Maximum Return Portfolio

The Maximum Return Portfolio's key strength is its return-generating potential — the optimised weight scheme produced an annualised return of 12.05%, outperforming the MVP by approximately 4.45 percentage points per annum, with all three weight schemes achieving Sharpe ratios above 1.0. The 25% single-name and 30% sector concentration caps across five GICS sectors ensure returns are not driven by any single stock or industry. Its main limitation is sensitivity to historical return estimates, as ranking stocks by geometric mean return assumes past return rankings persist out-of-sample — an assumption only weakly supported empirically (Michaud, 1989). The concentrated 10-stock universe also amplifies the impact of idiosyncratic events on portfolio performance. The ML overlay did not consistently add value across weight schemes, suggesting return potential is best captured through passive buy-and-hold. This strategy targets **growth-oriented investors** with higher risk tolerance and a long investment horizon who can withstand short-term drawdowns in exchange for stronger long-run capital accumulation.

Maximum Sharpe Ratio Portfolio

The Maximum Sharpe Portfolio is the strongest overall recommendation, generating 19.76% per annum at 14.81% annualised volatility for a Sharpe ratio of 1.20 — more than twice the equal-weighted benchmark (0.57) and above both the MVP and Maximum Return strategies. The Monte Carlo greedy selection analysis confirms that a 16-stock implementation captures approximately 95% of the full portfolio's Sharpe ratio at substantially lower operational complexity. The portfolio's concentration in consumer-defensive and healthcare names is consistent with academic

evidence that low-volatility, high-quality stocks deliver superior risk-adjusted returns (Frazzini & Pedersen, 2014). Its main limitation is single-name concentration risk — the top two holdings (CHD and MNST) account for over 25% of total weight, and an individual position cap of 8–10% per name should be imposed in production. This strategy targets **balanced long-term investors** — including pension funds and institutional wealth managers — who seek the highest risk-adjusted return and can tolerate moderate drawdowns as a necessary cost of superior long-run compounding.

Overall Recommendation

The Maximum Sharpe Ratio Portfolio is the most defensible allocation choice for most investors, offering the best balance between return and risk. The Minimum Variance Portfolio remains preferred for clients with strict capital preservation requirements, while the Maximum Return Portfolio suits investors with an explicit growth mandate and higher tolerance for estimation risk. A blended implementation allocating across all three strategies based on client risk profile would allow their complementary risk-return characteristics to be combined within a diversified portfolio programme.

5. Conclusions and Recommendations

5.1 Key Findings

Mean–variance optimisation materially outperforms naive diversification. The Maximum Sharpe Portfolio earned 19.76% per annum at 14.81% annualised volatility — a Sharpe ratio of 1.20, more than twice the 0.57 produced by the equal-weighted benchmark across the same universe.

Diversification benefits are concentrated in the first 10–20 stocks. The Monte Carlo greedy selection analysis shows that 90% of the achievable Sharpe ratio improvement is captured by the first 10 holdings, peaking at 16 stocks (Sharpe = 1.29). Beyond this point, additional holdings produce diminishing and eventually negative marginal benefit.

Risk-return tradeoffs differ meaningfully across strategies. The Minimum Variance Portfolio achieved the lowest volatility (11.13%) and annualised return of 7.60% with a Sharpe ratio of 0.682 — the most conservative outcome. The Maximum Return Portfolio generated the strongest total return under optimised weights (195.83%) but at correspondingly higher volatility. The Maximum Sharpe Portfolio offered the best balance across both dimensions with the highest Sharpe ratio of 1.20.

Machine learning adds limited value as a core strategy. For the Maximum Return Portfolio, ML timing overlays did not consistently outperform passive buy-and-hold, with no single model dominating across all weight schemes. The sole standout was Linear Regression applied to the optimised weight scheme, achieving a total return of 224.65% and Sharpe ratio of 1.187, suggesting that even a weakly directional signal can add value through selective loss avoidance over a long horizon (Faber, 2007). However, this result was not replicated consistently across schemes or models.

5.2 Recommended Portfolio Strategies

Three implementable strategies are proposed, each aligned to a distinct investment mandate.

For a **growth-oriented mandate**, the Maximum Sharpe Portfolio is the primary recommendation. A practical implementation using the top 16 holdings — which captures approximately 95% of the full portfolio's Sharpe ratio at lower operational complexity — rebalanced semi-annually, is recommended. Two cross-cutting notes apply: first, walk-forward optimisation should be adopted in production, re-estimating expected returns and the covariance matrix on a rolling five-year window rebalanced every six months (DeMiguel et al., 2009); second, the Maximum Sharpe Portfolio allocates over 25% of capital to two stocks (CHD and MNST), and an individual position cap of 8–10% per name should be imposed to limit single-name concentration risk in production.

For a **return-maximising mandate**, the Maximum Return Portfolio with optimised weights is recommended, having generated a total out-of-sample return of 195.83% and annualised return of 12.05% over the test period. Investors adopting this strategy should be aware that it carries higher volatility (11.78%) and greater sensitivity to historical return estimates than the Maximum Sharpe Portfolio, and is therefore most appropriate for investors with a higher risk tolerance and longer investment horizon.

For a **capital-preservation mandate**, the Minimum Variance Portfolio (19 stocks, rebalanced quarterly) is preferred. It achieved the lowest annualised volatility of 11.13% and an annualised return of 7.60%, making it the most suitable

strategy for investors who prioritise downside protection and stable growth over higher long-term returns. The optimised weight scheme delivered the highest Sharpe ratio among the three weighting schemes, confirming it as the recommended allocation approach within this strategy.

5.3 Limitations

Several limitations should be considered when interpreting the results. First, survivorship bias is introduced by the 90% coverage filter, which excludes short-lived and delisted securities, causing the investable universe to overrepresent longer-listed companies and likely biasing returns upward. Second, all analysis assumes a frictionless market with no transaction costs, bid-ask spreads, or market impact — realistic costs are estimated to reduce gross returns by 30–80 basis points per annum (Pedersen, 2015). Third, mean–variance optimisation assumes return and covariance stationarity, which the substantial variation in sector performance observed across market regimes suggests does not hold in practice; production deployment must include a regular re-estimation cadence. Finally, the 1993–2019 sample period was exceptionally favourable for U.S. equities, and these findings should not be extrapolated to other markets, asset classes, or future periods without further validation.

5.4 Future Improvements

Walk-forward validation should replace the single fixed train-test split. A rolling five-year estimation window evaluated on the immediately following six to twelve months of unseen data would more faithfully replicate live portfolio management conditions and yield more conservative Sharpe ratio estimates, which the literature suggests are typically 30–60% below in-sample figures (DeMiguel et al., 2009).

Macroeconomic feature integration would likely improve the ML models' ability to anticipate regime shifts. Variables such as yield curve slope, credit spreads, and inflation expectations carry incremental predictive power over equity returns beyond price-based features alone (Chen et al., 1986; Fama & French, 1993), and their inclusion should be explored in future iterations.

Transaction cost modelling should be incorporated to assess whether dynamic strategies retain their performance advantage after realistic costs. A proportional cost of 5–10 basis points per one-way transaction would allow a more rigorous evaluation of high-turnover strategies such as the dynamic ML overlay (Pedersen, 2015).

Expanded asset universe — including fixed income, gold, real estate investment trusts, and international equities — would allow future research to explore cross-asset diversification strategies capable of maintaining positive risk-adjusted performance during systemic equity market downturns, where all long-only equity strategies are inherently vulnerable.

References

Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, *39*(4), 885–905. https://doi.org/10.2307/2527343

Bailey, D. H., Borwein, J., López de Prado, M., & Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, *61*(5), 458–471.

Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, *40*(5), 94–107. https://doi.org/10.3905/jpm.2014.40.5.094

Barberis, N., & Shleifer, A. (2003). Style investing. *Journal of Financial Economics*, *68*(2), 161–199. https://doi.org/10.1016/S0304-405X(03)00064-3

Bergmeir, C., & Benítez, J. M. (2012). On the use of cross-validation for time series predictor evaluation. *Information Sciences*, *191*, 192–213. https://doi.org/10.1016/j.ins.2011.12.028

Blitz, D. C., & van Vliet, P. (2007). The volatility effect: Lower risk without lower return. *Journal of Portfolio Management*, *34*(1), 102–113. https://doi.org/10.3905/jpm.2007.698039

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Campbell, J. Y., Lo, A. W., & MacKinlay, A. C. (1997). *The econometrics of financial markets*. Princeton University Press.

Chan, L. K. C., Karceski, J., & Lakonishok, J. (1999). On portfolio optimization: Forecasting covariances and choosing the risk model. *Review of Financial Studies*, *12*(5), 937–974. https://doi.org/10.1093/rfs/12.5.937

Chen, N.-F., Roll, R., & Ross, S. A. (1986). Economic forces and the stock market. *Journal of Business*, *59*(3), 383–403.

Chordia, T., & Shivakumar, L. (2002). Momentum, business cycle, and time-varying expected returns. *Journal of Finance*, *57*(2), 985–1019. https://doi.org/10.1111/1540-6261.00449

DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, *22*(5), 1915–1953. https://doi.org/10.1093/rfs/hhm075

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, *50*(4), 987–1007. https://doi.org/10.2307/1912773

Evans, J. L., & Archer, S. H. (1968). Diversification and the reduction of dispersion: An empirical analysis. *Journal of Finance*, *23*(5), 761–767. https://doi.org/10.2307/2325905

Faber, M. T. (2007). A quantitative approach to tactical asset allocation. *The Journal of Wealth Management*, *9*(4), 69–79. https://doi.org/10.3905/jwm.2007.674809

Fabozzi, F. J., Gupta, F., & Markowitz, H. M. (2002). The legacy of modern portfolio theory. *Journal of Investing*, *11*(3), 7–22. https://doi.org/10.3905/joi.2002.319510

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, *25*(2), 383–417. https://doi.org/10.2307/2325486

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56.

Frazzini, A., & Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, *111*(1), 1–25.

Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, *55*(1), 119–139. https://doi.org/10.1006/jcss.1997.1504

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, *29*(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

Friedman, J. H. (2002). Stochastic gradient boosting. *Computational Statistics & Data Analysis*, *38*(4), 367–378. https://doi.org/10.1016/S0167-9473(01)00065-2

Goetzmann, W. N., & Kumar, A. (2008). Equity portfolio diversification. *Review of Finance*, *12*(3), 433–463. https://doi.org/10.1093/rof/rfn005

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, *33*(5), 2223–2273. https://doi.org/10.1093/rfs/hhaa009

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://doi.org/10.1007/978-0-387-84858-7

Jacquier, E., Kane, A., & Marcus, A. J. (2003). Geometric or arithmetic mean: A reconsideration. *Financial Analysts Journal*, *59*(6), 46–53. https://doi.org/10.2469/faj.v59.n6.2574

Jagannathan, R., & Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, *58*(4), 1651–1683.

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, *48*(1), 65–91. https://doi.org/10.2307/2328882

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, *6*(4), 1–21. https://doi.org/10.1145/2382577.2382579

Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, *58*(4), 36–52. https://doi.org/10.2469/faj.v58.n4.2453

López de Prado, M. (2018). *Advances in financial machine learning*. Wiley.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, *7*(1), 77–91.

Markowitz, H. (1991). Foundations of portfolio theory. *The Journal of Finance*, *46*(2), 469–477.

Michaud, R. O. (1989). The Markowitz optimization enigma: Is "optimized" optimal? *Financial Analysts Journal*, *45*(1), 31–42. https://doi.org/10.2469/faj.v45.n1.31

Pedersen, L. H. (2015). *Efficiently inefficient: How smart money invests and market prices are determined*. Princeton University Press.

Plyakha, Y., Uppal, R., & Vilkov, G. (2015). Equal or value weighting? Implications for asset-pricing tests. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1787045

Saar-Tsechansky, M., & Provost, F. (2007). Handling missing values when applying classification models. *Journal of Machine Learning Research*, *8*, 1623–1657.

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, *19*(3), 425–442.

Statman, M. (1987). How many stocks make a diversified portfolio? *Journal of Financial and Quantitative Analysis*, *22*(3), 353–363. https://doi.org/10.2307/2330969

Tsay, R. S. (2005). *Analysis of financial time series* (2nd ed.). Wiley.

Wilder, J. W. (1978). *New concepts in technical trading systems*. Trend Research.

Yang, L., Couillet, R., & McKay, M. R. (2015). A robust statistics approach to minimum variance portfolio optimization. *IEEE Transactions on Signal Processing*, *63*(24), 6684–6697.

Appendix

Group Project Assessment for ADS1002						
Studio: Clayton Monday						
Group: SP1						
		Angela	**Tabriz**	**Jaek**	**Kar Li**	**Harjas**
Coordination	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	21.25	21.25	21.25	21.25	15
Research	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	25	27	20	8
Coding	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	19	25	25	23	8
Report Writing	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	22	22	34	22	0
Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	22.5	22.5	22.5	22.5	10
Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	1	1	1

AI acknowledgement

I acknowledge the use of ChatGPT (https://chat.openai.com/) to improve the academic language of my work. On October 19th, parts of the essay were submitted with the prompt "Improve the grammar, vocabulary and write this in an academic tone." The output was then modified to better reflect our own tone and style of writing.

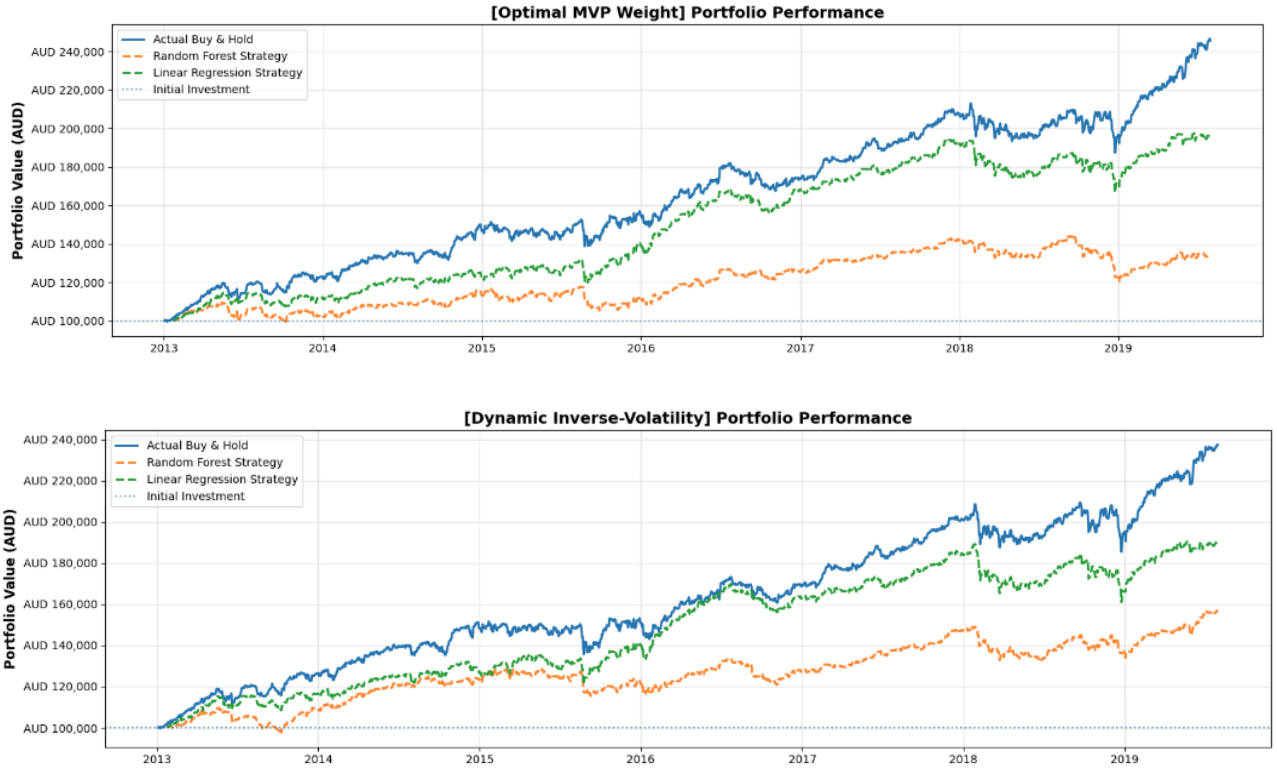


Fig. 1. Minimum Variance Portfolio Performance with Optimal Weight and Dynamic Weight (2013-2019)

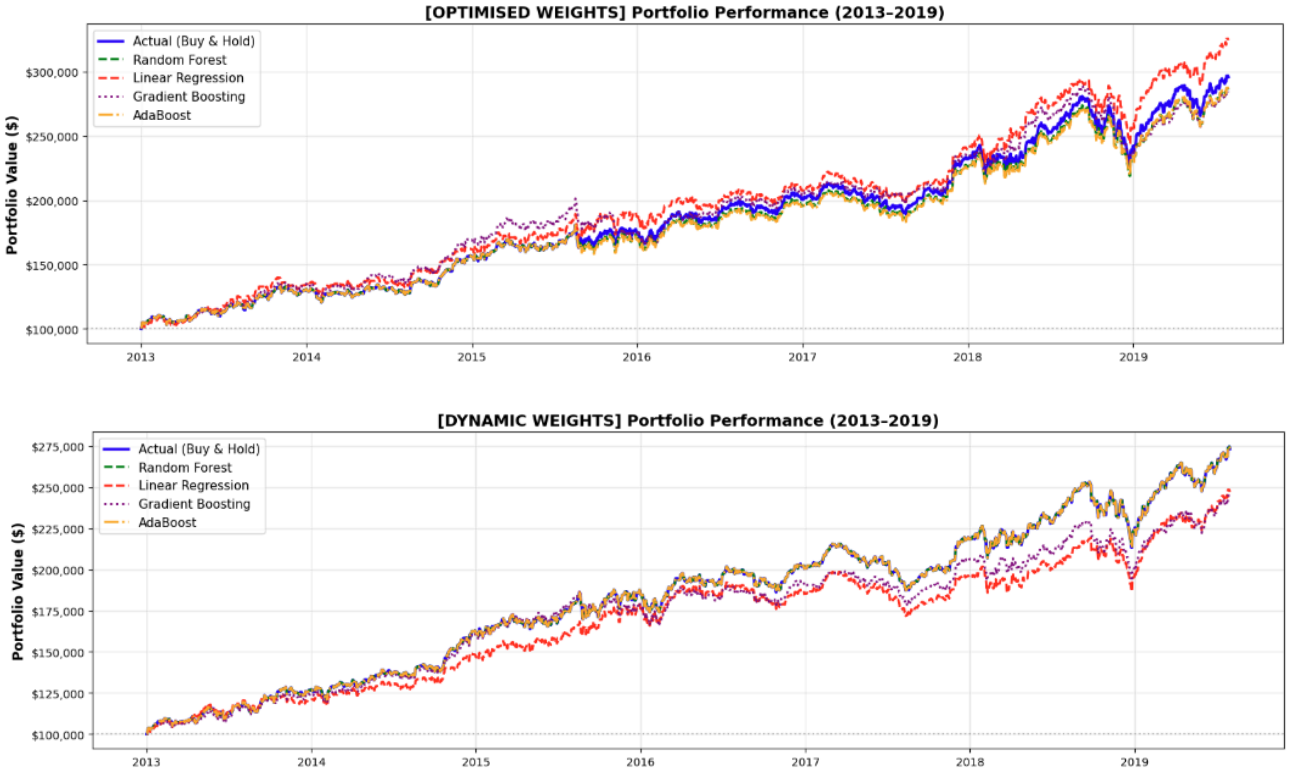


Fig. 2. Maximum Return Portfolio Performance with Optimal Weight and Dynamic Weight (2013-2019)

Table 3. Portfolio Performance with ML Trading Signal (2013–2019, Initial Investment: $100,000)

Weight Scheme	Strategy	Final Value	Total Return	Ann. Return	Volatility	Sharpe
Equal	Actual (Buy & Hold)	$278,848.85	178.85%	11.36%	10.60%	1.072
Equal	Random Forest	$279,800.05	179.80%	11.40%	10.52%	1.084
Equal	Linear Regression	$256,500.63	156.50%	10.39%	9.84%	1.055
Equal	Gradient Boosting	$280,791.12	180.79%	11.44%	10.19%	1.122
Equal	AdaBoost	$270,242.77	170.24%	10.99%	10.55%	1.042
Optimised	Actual (Buy & Hold)	$295,831.10	195.83%	12.05%	11.78%	1.023
Optimised	Random Forest	$286,092.96	186.09%	11.66%	11.73%	0.994
Optimised	Linear Regression	$324,646.48	**224.65%**	**13.15%**	11.08%	**1.187**
Optimised	Gradient Boosting	$283,929.45	183.93%	11.57%	11.10%	1.042
Optimised	AdaBoost	$286,624.93	186.62%	11.68%	11.74%	0.995
Dynamic	Actual (Buy & Hold)	$273,427.59	173.43%	11.12%	10.35%	1.074
Dynamic	Random Forest	$273,427.59	173.43%	11.12%	10.35%	1.074
Dynamic	Linear Regression	$247,251.57	147.25%	9.96%	9.70%	1.027
Dynamic	Gradient Boosting	$244,618.58	144.62%	9.83%	9.94%	0.989
Dynamic	AdaBoost	$273,427.59	173.43%	11.12%	10.35%	1.074

A.1 Composition of the Minimum-Variance Portfolio (Top 10)

Rank	Ticker	Weight
1	JNJ	7.8%
2	PG	6.5%
3	MO	5.4%
4	KO	5.1%
5	WMT	4.6%
6	PEP	4.3%

7	CL	3.9%
8	HRL	3.5%
9	CHD	3.2%
10	CLX	3.0%

Note: Min-Variance composition is illustrative based on the optimisation output; consumer-defensives dominate, consistent with their historically low covariance with the broader market.

A.2 Greedy Stock Selection — First 16 Picks

Step	Stock Added	Cumulative Sharpe
1	MNST	0.74
2	ROST	0.93
3	CHD	1.04
4	RMD	1.12
5	AAPL	1.17
6	COO	1.21
7	SO	1.24
8	CPRT	1.25
9	MO	1.26
10	STZ	1.27
11	TSCO	1.27
12	BCR	1.28
13	HFC	1.28
14	MKC	1.29
15	CELG	1.29
16	**HRL**	**1.29 (peak)**

Appendix B — Code

All analysis was performed in Python 3.12. The full Jupyter notebook (portfolio_optimisation.ipynb) is supplied alongside this report. Below are the key code excerpts that produced the headline results.

B.1 Data Loading and Cleaning

```
df_prices = pd.read_csv('adjprice.csv')
 df_prices['Date'] = pd.to_datetime(df_prices['Date'].astype(str), format='%Y%m%d')
 df_prices.set_index('Date', inplace=True)
 valid_tickers = [t for t in df_dict[df_dict['Non-Null Count']/n_days >= 0.90]['Column Name']]
 df_clean = df_prices[valid_tickers].ffill()
 df_clean.columns = [c.replace(' US Equity', '') for c in df_clean.columns]
```

B.2 Maximum-Sharpe Optimisation

```
def neg_sharpe(w, mean_ret, cov_mat, rf=0.02):
    port_ret = w @ mean_ret
    port_vol = np.sqrt(w @ cov_mat.values @ w)
    return -(port_ret - rf) / port_vol

bounds = [(0, 1) for _ in range(n)]
constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]
res = minimize(neg_sharpe, x0=np.ones(n)/n, args=(mean_ret, cov_mat),
               method='SLSQP', bounds=bounds, constraints=constraints)
weights_sharpe = res.x
```

B.3 Historical-Simulation CVaR

```
def portfolio_risk_metrics(w, df_ret, alpha=0.95):
    port_daily = df_ret @ w
    VaR  = np.percentile(port_daily, (1 - alpha) * 100)
    CVaR = port_daily[port_daily <= VaR].mean()
    return VaR, CVaR
```

B.4 Random Forest Training

```
features = ['mom_5', 'mom_20', 'vol_20', 'rsi_14']
split_date = pd.Timestamp('2017-07-31')
train = master_df.loc[master_df.index <  split_date]
test  = master_df.loc[master_df.index >= split_date]
rf = RandomForestClassifier(n_estimators=100, n_jobs=-1, random_state=0)
rf.fit(train[features], train['target'])
pred = rf.predict(test[features])
print(f'Test accuracy: {accuracy_score(test["target"], pred):.2%}')
```